Royal Devon University Healthcare NHS Foundation Trust

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Clinical Guideline for: **Code Red (Northern Services)**

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SUMMARY

This guideline outlines the process for the provision of emergency blood components when ED staff receive a pre-hospital alert for an adult patient with life-threatening injuries or illness.

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CODE RED (Northern ED)

Ring 2222

and state "Code Red ED Resus"

ED next actions

- Prepare resus bay
- Allocate two nurses to run the rapid infuser
- Allocate transfusion doctor

Which blood components will be issued?

- **Code Red Pack in boxes:**
  - x2 Group O D-negative red cells
  - x2 Group A or AB FFP
- **x2 further units of Emergency O red blood cells available in the Blood Fridge at Pathology Level 1**

If patient still bleeding

- **Follow Major Haemorrhage (MH) flowchart**

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement and adaptation to fit individual circumstances.

1.2 Following a pre-hospital alert, this Code Red guideline enables Emergency Department (ED) staff at the North Devon District Hospital to support transfusion to adult patients with life-threatening injuries or illness at the earliest opportunity upon their arrival to hospital.

1.3 Patients with life threatening haemorrhage require urgent replacement of lost circulating volume with blood component support.

1.4 Code Red provides a distinct and easily identifiable alert to activate the transfusion emergency response.

1.5 This guideline applies to individual patients but should this be required for multiple patients at the same time, this is the explicit responsibility of the ED Team Leader to liaise with transfusion staff for this unique circumstance or consider calling a major incident.

1.6 Code Red is for ED only.

2. BACKGROUND

2.1 The British Society for Haematology (BSH) guidance (2022) mandates a multidisciplinary approach involving the close working between laboratories and clinical departments enabling a timely, targeted approach to transfusion support for patients who are actively bleeding.

2.2 This Code Red guideline formalises the process to enable rapid provision of emergency blood components in validated blood component transport boxes to a bleeding patient through agreed communication channels between ED staff and the transfusion laboratory without escalation for approval.

2.3 The emphasis is on early recognition and treatment of patients who need immediate blood component resuscitation.

2.4 Communication and co-ordination are key to optimising the delivery of a safe and effective transfusion response and enabling better use of resources.

3. DEFINITIONS

3.1 Emergency Red Blood Cells (RBCs)

3.1.1 Group O red cell components that are not crossmatched against, or labelled for, a specific patient but are required to provide critical life-saving support based on clinical judgement of the severity of bleeding.

3.2 Emergency Fresh Frozen Plasma (FFP)

3.2.1 Group A or AB FFP components that are not labelled for a specific patient but are required and thawed to provide critical life-saving support for volume expansion and replacement of coagulant factors.

3.3 Cold Chain and traceability

3.3.1 Blood components must be kept at the required temperature at all times to prevent them becoming infected; infected blood components can cause fatal reactions in patients.

3.3.2 The cold chain is a mandatory requirement which starts from the delivery of the component from the blood centre to the time that the component is transfused or is disposed of.

3.3.3 An unbroken cold chain is an uninterrupted series of storage and distribution activities which maintain a given temperature range.

3.3.4 The blood component transport boxes are validated for storage up to 4 hours as long as the seal is not broken and the box is not opened.

3.3.5 A temperature logger is contained within the box for laboratory use. If the box is opened and components used for transfusion please ensure that the logger is left in the box and returned to the laboratory.

3.3.6 Traceability of blood components is a legal requirement. BloodTrack Tx must be used to confirm traceability to the patient as part of the transfusion pathway.

4. ACTIVATING CODE RED

4.1 Code Red can be activated by the ED Team Leader (ED Senior Doctor/Consultant) based on the information provided by the pre-hospital alert.

4.2 Code Red can be activated for pre-hospital alert patients with history compatible with significant haemorrhage (eg trauma, UGIB) with evidence of abnormal physiology.

5. ACTIONS FOR ED STAFF

5.1 ED staff to ring 2222 to request Code Red ED Resus if the details provided on the pre-hospital alert meet 4.2.

5.2 Prepare the resus bay.

5.3 The rapid infuser is primed with 0.9% sodium chloride in readiness for the patient arrival.

5.4 Blood samples for an ABO group and antibody screen should be taken as early as possible whilst securing IV access and ideally before emergency components are transfused.

5.5 ED Team Leader to decide if the Code Red components are to be administered immediately or held in the sealed boxes and returned to the transfusion laboratory ASAP. Note: seal is NOT to be broken in anticipation of use to avoid unnecessary waste.

5.6 Use BloodTrack Tx to confirm traceability and record on EPIC the blood components transfused.

5.7 Update transfusion staff of the situation as soon as practical and ED Team Leader to initiate the major haemorrhage guidance if appropriate.

6. ACTIONS FOR SWITCHBOARD

6.1 Initiate Code Red.

6.2 Code Red team include transfusion laboratory staff (bleep 045), the Chargehand Porter (bleep 506) and Transfusion Nurse (bleep 280).

7. ACTIONS FOR LABORATORY STAFF (Bleep 045 holder)

7.1 The transfusion laboratory staff will be alerted by Switchboard when Code Red is activated.

7.2 Complete the Code Red telephone record TRLABFOR-0172.

7.3 Prepare x2 units of emergency O D-negative red blood cells (RBCs) and x2 emergency group A or AB fresh frozen plasma (FFP) as per SOP (T-SOP-23).

7.4 Put the RBCs into a transport box for collection by the porter.

7.5 Follow the Code Red telephone form for instructions relating to the temperature logger.

7.6 Use BloodTrack Manager to ‘move out’ the components into a ‘cooler’ for 4-hour storage time.

7.7 When the FFP is ready, wait for the porter to arrive before placing them into a transport box to ensure maximum storage time.

7.8 Process transfusion sample(s) without delay once arrived in the laboratory.

7.9 Manage blood stocks with consideration of placing an order with NHSBT or Exeter if appropriate (e.g. group O red cell stocks are low).

7.10 Await update from ED staff and proceed to major haemorrhage guidance if instructed.

8. ACTIONS FOR THE CHARGEHAND PORTER / PORTER

8.1 The Chargehand Porter will be alerted by Switchboard when Code Red is activated.

8.2 The Chargehand Porter will go the transfusion laboratory or delegate another porter to collect the box of emergency RBCs from the transfusion lab staff.

8.3 The Porter will take the box of emergency RBCs to ED Resus.

8.4 If the patient has arrived, the Porter to wait for the transfusion sample(s) to take to the transfusion lab and return to ED Resus with the box of emergency FFP.

8.5 If the patient has not arrived, the Porter to collect the box of emergency FFP from the transfusion lab staff and return to ED Resus.

8.6 The Porter to await further instructions from ED staff.

9. EMERGENCY BLOOD COMPONENTS

9.1 Blood components are only used on the instruction of the ED Team Leader.

9.2 The transfusion sample(s) should be collected before emergency blood components are transfused.

9.3 The Transfuse Orders must be placed on EPIC to record the components transfused.

9.4 The pre-transfusion checks and visual checks should be carried out on all blood components prior to administration and BloodTrack Tx used to trace the components to the patient.

9.5 If the emergency components are not required, ED staff to arrange return of both boxes to the transfusion laboratory. If the boxes remain sealed, the blood components can be returned to stock. If the boxes are opened any remaining components not used will be disposed of by transfusion staff as the cold chain will have been compromised.

10. MONITORING COMPLIANCE WITH THIS GUIDELINE

10.1 Monitoring the implementation, effectiveness and compliance with these guidelines will be the responsibility of the Hospital Transfusion Team (HTT).

11. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

- Blood Transfusion Policy
- Lab SOP for the provision of emergency blood products T-SOP-23
- Stanworth et al (2022) Haematological management of major haemorrhage: a British Society for Haematology Guideline